CR CSE 14

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2017
Department of Computer Science and Engineering
CSE 4215
Information Security and Control

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define cryptosystem. Discuss about the ingredients of a cryptosystem. (08)
   b) Through Shannon characteristics, discuss the characteristics of good cipher. Compare between block and stream ciphers. (10)
   c) Define chosen-plaintext attack. Discuss a specific key exchange protocol to distribute a key. (10)
   d) Define cryptographic hash function. Discuss some properties of it. (07)

2. a) Explain the mechanism of triple DES encryption. Using a block diagram explain AES algorithm. (13)
   b) Discuss properties of digital signature. Explain Elgamal digital signature algorithm with an example. (12)
   c) A computer network can become vulnerable. How? Explain the reasons. (10)

3. a) In RSA, how $ed = 1$? On same modulus using RSA, how multiple parties can encrypt data? (10)
   b) Define sensitive data. Discuss the factors that can make data sensitive. (10)
   c) Explain why and how digital signature and public key cryptography can be combined? (08)
   d) Explain the steps of signing documents with public key cryptography and one way hash functions. (07)

4. a) Discuss the dimension of reliability and integrity of database. What are the requirements of database security? Explain them. (10)
   b) Define 'primitive element' with example. How it is associated with Elgamal cryptosystem? (10)
   c) Define 'confusion' and 'diffusion'. Give an overview of DES encryption technique. (08)
   d) How is trade secret protection vanished by 'reverse engineering'? Explain. (07)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Consider the code snippet shown in below: (07)

```
int main()
{
  int pass = 0;
  char buff[3];
  printf("\n Enter the password:\n");
  gets(buff);
  if(strcmp(buff,"cse"))
  {
   printf("\n Wrong password\n");
  }
  else
  {
   printf("\n Correct password\n");
   pass = 1;
  }
if(pass)
 {
  /*Give admin priviledge to the  user*/
  printf("\n Admin priviledge given to the user\n");
  }
  return 0;
}
```

Explain how buffer overflow attack can be utilized to get admin privilege without even knowing the actual password.

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b) With appropriate stack layout explain how buffer overflow attack can be used to modify return address of a function. Design an input payload for the code snnipet shown in the following to transfer the execution to secret function. (15)

```
void secretFunction()
{
printf("Can't be called");
}
void test()
{
char buff[10];
gets(buff);
}
int main()
{
test();
return 0;
}
```

```
Dump of assembler code for function
secretFunction:
0x004014d0<+0>: push %ebp
0x004014d1<+1>: mov %esp, %ebp
.
.
.
```

c) Explain how buffer overflow can be exploited to execute attacker's provided code. (06)
d) Explain integer overflow attack with an example. (07)

6. a) Discuss some prevention mechanisms that are used against buffer overflow attack. (12)
b) Explain why the code snnipet shown in the below is vulnerable to format string attack. (06)

```
int main (int argc, char *argv[])
{
char *secret = "test";
char *secret2 = "test2";
printf(argv[1]);
return 0;
}
```

c) Explain how format string vulnerability can be utilized for: (i) Crashing a program (ii) Viewing the stack (iii) Viewing memory at any location (iv) Writing an integer to nearly any location in process memory. (17)

7. a) Suppose, you want to perform brute force attack on a system which allows password length between 5 to 8 characters limited to character set:{'a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j', '1', '2', '3', '4'}. Now, write a pseudocode to generate all possible passwords for your brute force attack. (12)
b) Explain how the tapjacking attack works. (05)
c) What are covert channels? Explain how covert channels can be created. (12)
d) Write short notes on: (i) Trapdoors (ii) Keystroke logging (06)

8. a) Explain how Dynamic Data Exchange (DDE) and macro in Microsoft Office can be exploited to run malicious code. (05)
b) Explain the working principle of ransomware. (05)
c) Explain the terms: i) denial-of-service (DoS) attack and ii) distributed DoS attack. (08)
d) Discuss the threats to E-mail. Also discuss how to defend against it. (07)
e) What is meant by malleability? Show that both RSA and Elgamal are malleable cryptosystem. (10)

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